

GUIDELINES FOR NVIDIA DLSS

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NVIDIA DLSS

NVIDIA DLSS is a suite of neural rendering technologies that delivers best-in-class image quality, performance, and responsiveness. The suite comprises DLSS 3D-Guided Neural Rendering, DLSS Dynamic Multi Frame Generation, DLSS Frame Generation, DLSS Super Resolution, DLSS Ray Reconstruction, NVIDIA DLAA, NVIDIA Reflex Low Latency. To deliver this intended gaming experience, it is important for users to be able to easily toggle all components of DLSS with one global switch.

- ***DLSS 3D-Guided Neural Rendering*** uses AI tuned by the developer to enhance graphics with lifelike lighting and materials. DLSS Neural Rendering is available for GeForce RTX 50 Series GPUs.
- ***DLSS Multi Frame Generation*** uses AI to boost frame rates by generating up to 5 additional high-quality frames, all while optimizing responsiveness with NVIDIA Reflex. Also includes Dynamic Multi Frame Generation, which dynamically adjusts frame generation multiplier to achieve target frame rate. DLSS Multi Frame Generation is available for GeForce RTX 50 Series GPUs.
- ***DLSS Frame Generation*** uses AI to boost frame rates by generating one additional high-quality frame, all while optimizing responsiveness with NVIDIA Reflex. DLSS Frame Generation requires a GeForce RTX 40 Series GPU or higher and a Reflex SDK integration.
- ***DLSS Super Resolution*** boosts frame rates by rendering fewer pixels and using AI to reconstruct high resolution frames. This feature is available for all RTX GPUs.
- ***DLSS Ray Reconstruction*** uses AI to enhance image quality and generate additional pixels for intensive ray-traced scenes. Ray Reconstruction replaces hand-tuned denoisers with an NVIDIA supercomputer-trained AI network that generates higher-quality pixels in between sampled rays. This feature is available for all RTX GPUs and requires Super Resolution to be enabled.
- ***NVIDIA DLAA*** maximizes image quality through an AI-based anti-aliasing technique. DLAA uses the same Super Resolution technology developed for DLSS, reconstructing a native resolution image to maximize image quality. This feature is available for all RTX GPUs.

- ***NVIDIA Reflex*** helps DLSS Multi Frame Generation and Frame Generation achieve maximum performance at the lowest latency. It accomplishes this by synchronizing the CPU frame submission with GPU processing for just in time rendering.

UI Tooltip or Setting Description

- ***NVIDIA DLSS:*** NVIDIA DLSS uses AI to boost frame rates while maintaining great image quality and responsiveness. This feature requires a GeForce RTX graphics card.
    - ***Neural Rendering:*** DLSS Neural Rendering uses AI to enhance graphics with lifelike lighting and materials. DLSS Neural Rendering requires a GeForce RTX 50 Series GPU.
    - ***Frame Generation:*** Frame Generation boosts frame rates by using AI to render additional frames. 2X is supported on a GeForce RTX 40 Series GPU or higher, while Fixed and Dynamic through 6x are supported on a GeForce RTX 50 Series GPU.
    - ***Super Resolution:*** Super Resolution boosts frame rates by rendering fewer pixels and using AI to output high resolution frames. This feature requires a GeForce RTX GPU.
    - ***Ray Reconstruction:*** Ray Reconstruction enhances image quality by generating high-quality pixels between sampled rays for intensive ray traced content. This feature requires a GeForce RTX GPU.
    - ***DLAA:*** DLAA is an AI-based anti-aliasing mode to improve image quality. This feature requires a GeForce RTX GPU.
    - **NVIDIA Reflex Low Latency -** NVIDIA Reflex reduces system latency and increases PC responsiveness.

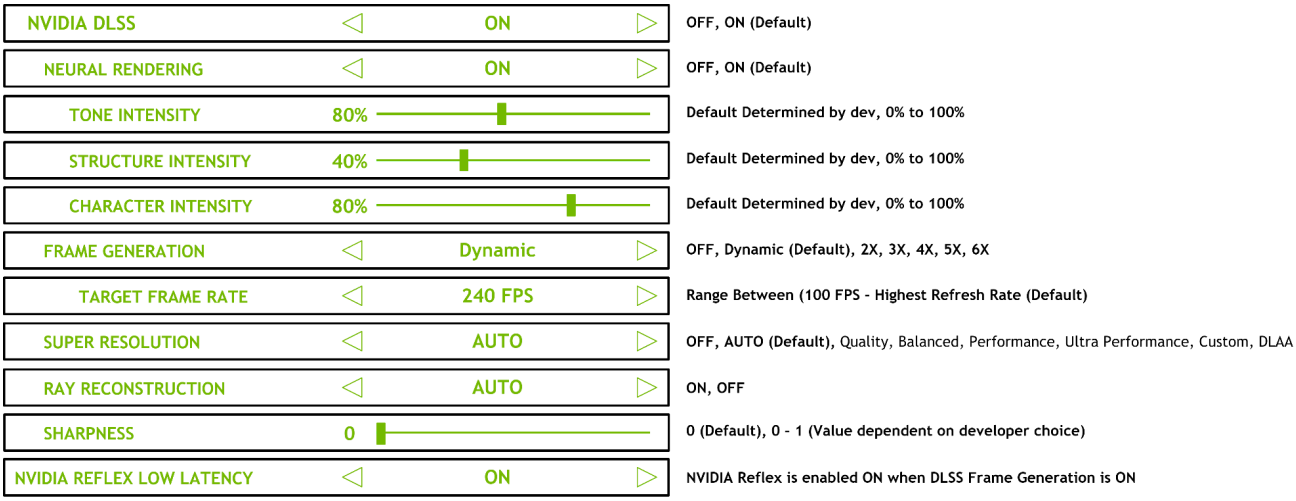
UI Menu Options

When DLSS Frame Generation is ON, please enable NVIDIA Reflex Low Latency.

- DLSS Frame Generation 2x is available for RTX 40 Series RTX graphics cards or higher.
- DLSS Frame Generation Dynamic, 3x - 6x are available for RTX 50 Series RTX graphics cards.

When DLSS Super Resolution is ON, please disable scaling technologies (Resolution Scaling, NVIDIA Image Scaling), and Anti-Aliasing. DLSS Super Resolution is available for all RTX graphics cards.

UI: DLSS ON



Developer UI Checklist

- **DLSS Feature Support**
  - We **strongly** recommend a global DLSS Feature toggle, which enables DLSS Frame Generation, DLSS Super Resolution, Ray Reconstruction, and NVIDIA Reflex Low Latency mode. This is the easiest way for end users to get the targeted performance, image quality, and latency benefits of DLSS.
  - Please make sure NVIDIA DLSS is **disabled** when on unsupported hardware or drivers.
  - Please make sure PCL Markers are always enabled on all vendor GPUs, whether Reflex is toggled on or off. This allows all gamers to accurately measure their PC latency, even on non NVIDIA GPUs.
- **DLSS 3D-Guided Neural Rendering**
  - We recommend enabling DLSS Super Resolution and DLSS Frame Generation when DLSS 3D-Guided Neural Rendering is enabled.
- **DLSS Frame Generation**
  - DLSS "Auto" is available for RTX 40 Series RTX graphics cards and set to 2x by default.
  - DLSS "Dynamic" is available for RTX 50 Series RTX graphics cards and set to 6x by default.
  - DLSS Frame Generation **requires** NVIDIA Reflex Low Latency integration and enablement. Users should not be able to disable NVIDIA Reflex Low Latency when Frame Generation is ON.
  - DLSS Frame Generation **requires** Windows OS "Hardware-accelerated GPU Scheduling" and fails to initialize if not enabled. If this occurs, the game should notify the user to enable this feature.
- **DLSS Super Resolution**
  - The DLSS mode titled "Auto" should be the first option in the UI after Off and be enabled by default when NVIDIA RTX hardware is detected. The Auto mode isn't itself a true mode and should select the appropriate default mode from the table below depending on the current output resolution.
  - When the UI shows the DLSS modes horizontally or in a left-right scrolling list, the order should be: Off, Auto, Quality, Balanced, Performance, Ultra-Performance, DLAA.
  - When the UI shows the DLSS modes vertically or in an up-down scrolling list, the order should be:
    1. Off
    2. Auto

3. Quality
4. Balanced
5. Performance
6. Ultra-Performance
7. DLAA

- When DLSS Super Resolution is turned on, **make sure the anti-aliasing settings are disabled** (both UI, as itself)
- When DLSS Super Resolution is turned on, **make sure the resolution scale settings are disabled** (UI disables, the application uses render target size from DLSS optimal settings)

- **DLSS Ray Reconstruction**
  - The DLSS mode titled "Auto" should be the first option in the UI after Off, and be enabled by default when NVIDIA RTX hardware is detected, DLSS Super Resolution is enabled, and when Ray Tracing content is enabled.
  - **Requires DLSS Super Resolution to be ON**. Disable when Super Resolution is not enabled.
  - Requires Ray Tracing to be ON. Disable when Ray Tracing is not enabled.
- **Sharpness Slider**
  - Sharpness slider is a requirement for DLSS integration and the preferred integration is NVIDIA Image Scaling (please see below). When sharpness is on for DLSS, make sure other sharpening features are disabled.

DLSS 3D-Guided Neural Rendering Options

Mode	Description	Type	GPU Requirements
Off	Disables DLSS Neural Rendering	Toggle	N/A
ON	DLSS Neural Rendering uses AI tuned by developers to enhance graphics with lifelike lighting and materials. DLSS Neural Rendering requires a GeForce RTX 50 Series graphics card.	Toggle, Default ON	RTX 50 Series RTX GPUs
TONE INTENSITY (OPTIONAL)	Adjust the intensity of DLSS Neural Rendering on lightness, darkness, or color.	Slider (0-100%)	RTX 50 Series RTX GPUs
STRUCTURE INTENSITY (OPTIONAL)	Adjust the intensity of Neural Rendering on details, shadows, and materials.	Slider (0-100%)	RTX 50 Series RTX GPUs
CHARACTER INTENSITY (OPTIONAL)	Adjust the intensity of Neural Rendering on characters.	Slider (0-100%)	RTX 50 Series RTX GPUs

Note: Character is an optional feature to expose if a control mask is not used by the developer.

DLSS Multi Frame and DLSS Frame Generation Options

Mode	Description	Resolution Support	GPU Requirements
Off	Turn DLSS Frame Generation off.	N/A	N/A
Auto	Enables Frame Generation as needed to 2X to boost performance	ALL RESOLUTIONS	RTX 40 Series RTX GPUs ONLY
Dynamic	Dynamically enables Multi Frame Generation modes to target FPS.	ALL RESOLUTIONS	RTX 50 Series RTX GPUs
Target Frame Rate (OPTIONAL)	Optional control for Dynamic for 120Hz+ monitors. Sets target frame rate between 100 FPS to highest refresh rate (Default value) at current resolution. ie. [100 FPS - 480 FPS]	ALL RESOLUTIONS	RTX 50 Series RTX GPUs
2X	The total number of frames is doubled, with one rendered and one generated frame.	ALL RESOLUTIONS	RTX 40+ Series RTX GPUs
3X	Generates two additional frames for one rendered for a total of three frames.	ALL RESOLUTIONS	RTX 50 Series RTX GPUs
4X	Generates three additional frames for one rendered for a total of four frames.	ALL RESOLUTIONS	RTX 50 Series RTX GPUs
5X	Generates four additional frames for one rendered for a total of five frames.	ALL RESOLUTIONS	RTX 50 Series RTX GPUs
6X	Generates five additional frames for one rendered for a total of six frames.	ALL RESOLUTIONS	RTX 50 Series RTX GPUs

DLSS Super Resolution Options

Mode	Description	Resolution Support	GPU Support
OFF	Turns DLSS Super Resolution off.	N/A	N/A
AUTO	Selects the best DLSS Mode for the current output resolution.	ALL RESOLUTIONS	ALL RTX GPUs
QUALITY	Offers higher image quality than balanced mode.	ALL RESOLUTIONS	ALL RTX GPUs
BALANCED	Offers both optimized performance and image quality.	ALL RESOLUTIONS	ALL RTX GPUs

PERFORMANCE	Offers a higher performance boost than balanced mode.	ALL RESOLUTIONS	ALL RTX GPUs
ULTRA PERFORMANCE	Offers the highest performance boost.	ALL RESOLUTIONS	ALL RTX GPUs
DLAA	Recommended placement for DLAA.	ALL RESOLUTIONS	ALL RTX GPUs

These are the recommended DLSS settings based on output resolution. This is automatically set when user sets to 'AUTO"

Default DLSS Mode	Output (Resolution)	Output (Megapixels)
Disabled	Below 1920x1080	Below 2.03
Quality mode	Equal to 1920x1080, equal or below 2560x1440	Up to 3.68
Performance mode	Greater than 2560x1440, equal or below 3840x2160	3.69 - 8.29
Ultra Performance mode	3840x2160 and higher (e.g. 5120x2880 and 7680x4320)	3.69+

DLSS Ray Reconstruction Options

Mode	Description	Resolution Support	GPU Support
OFF	Turns Ray Reconstruction OFF	N/A	N/A
AUTO	Automatically enables Ray Reconstruction when Ray Tracing is ON, or disables it when Ray Tracing is OFF	ALL RESOLUTIONS	ALL RTX GPUs

DLSS Super Resolution & Dynamic Resolution Systems

As detailed in the DLSS Programming Guide, DLSS Super Resolution can support dynamically varying input sizes if the renderer has a Dynamic Resolution System (DRS).

If DRS is enabled, the game UI should:

1. Present only two options for DLSS Super Resolution: "Off" and "On"; or
2. If the UI system does not allow the DLSS Super Resolution options to change, disable (hide or gray out) all DLSS Modes and only allow the user to select "Off" or "Auto".

NVIDIA REFLEX

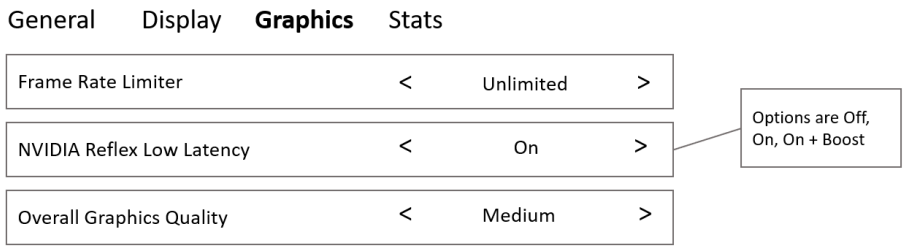
The Reflex SDK is an easy-to-integrate set of NVAPIs that provide both system latency optimization and measurement capabilities, giving players a more responsive feel in-game. The SDK strives to keep players at the lowest possible latency by dynamically adjusting submission timing of rendering work to the GPU so that they are processed just-in-time. The SDK includes the ability to measure PC Latency and automatically configure Reflex Analyzer which helps gamers optimize settings. Reflex also helps DLSS Frame Generation achieve maximum performance in addition to reducing latency.

UI Tooltip or Setting Description

- **NVIDIA Reflex Low Latency -** *NVIDIA Reflex reduces system latency and increases PC responsiveness.*

UI Menu Options

Low Latency Mode - "NVIDIA Reflex Low Latency"



Note: Reflex is required for DLSS Frame Generation, and will automatically turn on when DLSS Frame Generation is enabled. Reflex does not currently support other frame generation technologies (e.g. FSR), and should therefore be unavailable when these are enabled.

Settings Options

NVIDIA Reflex Low Latency

Mode	Description	Driver Support	GPU Support
Off	Low Latency mode is disabled.	N/A	N/A
On	Low Latency mode is now enabled and optimizes system latency.	R455+	Maxwell GPUs+ (Pascal+ for VK)
On+ Boost	Low Latency mode is now enabled and optimizes system latency. In Boost mode, NVIDIA Reflex will save slightly more latency and attempt to optimize latency in CPU bound cases. This option can slightly decrease FPS and increase GPU power draw.	R455+	Maxwell GPUs+ (Pascal+ for VK)

Settings Defaults

PC Latency is supported for all IHVs and should be **enabled at all times**. No UI setting is required

Notice

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